

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 February 2026
Team ID	LTVIP2026TMIDS88469
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Functional Requirements: Heart Disease Analysis

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Collect patient medical records dataset (age, gender, symptoms, vitals) Collect clinical test results (cholesterol, blood pressure, ECG, glucose, etc.) Collect target label (heart disease presence / absence)
FR-2	Data Cleaning & Processing	Clean raw data (remove duplicates, handle missing values) Transform and aggregate data for analysis (age groups, risk categories)
FR-3	Data Storage	Store raw dataset securely in MySQL Store cleaned and processed dataset in structured tables
FR-4	Data Visualization & Analysis	Build interactive dashboards in Tableau Visualize risk factors (age, cholesterol, BP, diabetes, smoking, etc.) Provide downloadable reports / dashboard screenshots for documentation

FR-5	Alerts & Insights Delivery	Display dashboards and key insights through Flask web application Embed Tableau dashboard link inside Flask
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Non-Functional Requirements: Heart Disease Analysis

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboards and Flask web pages should be easy to use and simple to understand for doctors and analysts.
NFR-2	Security	Patient health data must be stored securely in MySQL and should be accessed only by authorized users.
NFR-3	Reliability	The system should provide accurate and consistent results without dashboard errors or wrong insights.
NFR-4	Performance	Dashboards and visualizations should load quickly (within 5 seconds) for normal datasets.
NFR-5	Availability	The system should be available most of the time during working hours for analysis and reporting..
NFR-6	Scalability	The system should support adding more patient records and new medical factors in the future without issues.